Rupali Bhatnagar

Senior Software Engineer · AI-Assisted Engineering

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Senior engineer with **9+ years** across backend platforms, data pipelines and product engineering, six of them at **ServiceNow** building Security Operations features used by enterprise customers. Since 2025, independent engineering with AI agents in the loop: **Claude Code plugins, hooks and skills**, MCP, schema-constrained LLM output, eval fixtures drawn from real failures, and human-in-the-loop by construction — tools propose and explain, a person decides.

Work Experience

Independent	Software Consultant	Jun'25 – present

Platform and AI-tooling engineering for small businesses; building open-source developer tools alongside client work.

- **propagate** · Claude Code plugin & CLI, Node — open-source tool that keeps files which must agree across many repositories from silently drifting: **1,200+ declared file couplings**, drift derived from file content on demand rather than a stored baseline. Pre-tool **hooks** put the matching known hazard in front of an agent before it runs a risky command. Never edits a downstream file — it explains, a person approves, the decision lands in an append-only ledger. Every detector ships a selftest proving it can fire. **v0.15.41**, released continuously; **283 test files** — github.com/Rupali59/propagate.
- **Manuscript** · Obsidian plugin · TypeScript · Postgres — Claude-assisted editorial review and publishing: findings tracked in a Postgres review ledger, an author approves or rejects each one, and the agent side ships as a **Claude Code plugin with a hook and a skill** so every request carries a stable, versioned instruction set. **1.0.0 released** (private); **210 test files**; second contributor onboarding.
- **Legacy desktop software revival** · Windows · C# · reverse engineering — revived a discontinued astrology product whose vendor is unreachable: replaced its dead Flash dependency with a stub control, built an installer verified end to end on clean Windows 11 ARM, decoded its proprietary chart format (**350/350 round-trip**) and migrated its database to SQLite. Re-implemented the calculation engine in C#, validated against the original's output, as the spec for a **Flutter rebuild for iOS, Android, macOS and Windows**.
- **Motherboard** · Go 1.25 · Docker — multi-tenant API gateway and plugin platform, **21 Go modules**: client services register as plugins and publish their routes; tenant → authentication → RBAC path working end to end, **fail-closed entitlement gate**, scoped per-request tokens, notification hub live on Telegram.
- **Workflow engine cutover** · Express · Next.js · BullMQ — for a service-business CRM with database-driven, runtime-editable RBAC: a six-phase cutover that deleted the legacy passes only after freezing their outcomes as a **golden test** — surfacing five defects that existed only because the new graph had never executed. Test suite **104s → 28s**.
- **LLM classification pipeline** — Gemini under enforced response schemas at `temperature: 0`, determinism from cache-on-first-write rather than the model, keyword-classifier fallback so a provider outage degrades quality instead of stopping ingestion; model-proposed sources existence-checked against the live API.
- **Trading research** — contributed the research layer (edge estimators, hypothesis framework) and a design-drift ratchet with canaries to a team's algorithmic trading system.
- **Currently completing:** Manuscript's next release · the cross-platform Flutter suite (design approved) · Motherboard platform v1 · a release gate for propagate.
- **Also built:** four production portfolio sites (Next.js, Playwright, WCAG 2.2 AA gates; a video codec ladder 9.9 MB → 2.3 MB), a Shopify storefront with embedded Remix admin, a FastAPI astronomical-calculation service (**965 tests**), a classical-text ingestion pipeline (**92,166 verses**), and recovery of **154 unreachable git commits** from a live repository.

ServiceNow	Sr. Software Engineer (Security Operations)	Mar'22 – May'25

- **Exception Management — Approval Rule Engine** — multi-level, dynamic approval engine on Flow Designer for vulnerability deferment and exception requests across differing enterprise policies; shipped in **v15.0.2**. **Approval time down 40%.**
- **Background Job Processing Framework** — architected multithreaded execution for long-running tasks with status tracking, cancellation, retry and dead-letter queuing; shipped in **v18.2.5**. **60% efficacy improvement** in some use cases, **1M+ tasks/month**.
- **Penetration Testing Request Catalog** — automated intake, triage and routing across **50+ subsystems**, cutting manual coordination by **30%+**.
- **Real-Time Data Aggregation Engine** — near real-time multi-dimensional analytics over **100K+ records in an 8-hour window**.

ServiceNow	Software Engineer (Security Operations)	Apr'19 – Mar'22

- **Assignment Rule Engine** — cache-optimised routing of vulnerable items to remediation groups; **1M+ vulnerable items daily**.
- **ML Assignment Recommendations** — classification models on Predictive Intelligence with a reviewer-facing interface; shipped in **v12.1.4**. **Allocation accuracy up 20%.**
- **Vulnerability Source Integration** — ETL from Veracode, Qualys, Tenable and Microsoft TVM standardising **500K+ records/day** into a common model.
- **CISO & Vulnerability Dashboards** — led design and metrics; used by **50+ organisations**. **POA&M Generator** (NIST SP 800-53 / FedRAMP) — hackathon build adopted by the Now on Now team.

Darwinbox Digital Solutions	Software Developer	Aug'18 – Mar'19

- Audit-trail microservice listening to **MongoDB oplogs** and publishing to **AWS SQS**, with a Node.js processor diffing documents into human-readable audit entries.

NowFloats	Software Developer	Jun'17 – Jul'18

- Customer-relationship surface of the Boost merchant portal (C#/ASP.NET) serving **50,000+ merchant websites**; escrow-backed purchase flow and order management; appointment scheduling across verticals.
- Built **SPIDER**, an ML module predicting sales revenue per representative with up to **84% accuracy** — hackathon winner.

Finomena (wound down 2017)	Software Developer	Feb'17 – May'17

- Lending dashboards in Laravel scaling to **8,000+ loan applications/month**, and an SMS categorisation service feeding alternative underwriting for borrowers with no credit history.

Babajob.com (acquired by Quikr)	Software Engineer	May'16 – Jan'17

- Built **TelephonySystem** in **Go**: the missed-call job channel where a feature-phone user built a profile by ringing a number — **10,000+ calls and 20,000+ SMS/day** on RabbitMQ, Redis and MongoDB.
- Wired it into an automated matching pipeline over **Elasticsearch** that sent shortlisted jobs by SMS.

Technical Skills

AI & Agent Tooling	Claude Code plugins, hooks and skills; MCP; schema-constrained LLM output; eval fixtures and regression suites; graceful degradation; human-in-the-loop design; Gemini, Predictive Intelligence
Languages	Go, TypeScript, JavaScript, Python, C#, PHP, Java
Frameworks & Data	Next.js, React, Node.js, Express, FastAPI, Remix, Laravel · MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch · RabbitMQ, BullMQ, AWS SQS
Cloud & DevOps	AWS, Vercel, Fly.io, Docker, GitHub Actions, Jenkins, Linux

Education

M.Tech, 2016	Computer Science & Engineering (Pattern Recognition & Machine Learning), **NIT Goa** — 8.42 CGPA. Dissertation: human activity recognition with an HMM-based intermediate matching kernel.
B.E., 2014	Computer Science & Engineering, **LNCT Bhopal (RGPV)** — 8.58 CGPA, Honours. Chancellor's Scholarship (top 10 statewide), 2012 and 2013.

Certifications & Achievements

AI — 2026	**Anthropic:** Claude Code in Action; Claude Platform 101; Introduction to Claude Cowork. **Google:** Gemini Enterprise Agent Ready; Google Cloud Innovator; AI Fundamentals. credly.com/users/rupali-bhatnagar-59
Achievements	**Geektrust GOLD** (95.8, Readability and Maintainability at ADVANCED) · Winner, NowFloats Hackathon 2018 · Silver Medal, Persistent Systems Programming Contest 2013